ENGINEERING ELECTIVE · CLOUD TRACK

Linux Admin & Networking

Process Management, Cron Scheduling & Networking Fundamentals

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```
imran@ubuntu:~$ uptime
up 12 days, load 0.14
imran@ubuntu:~$ whoami
imran
imran@ubuntu:~$ hostname
cloud-node-01
```

SESSION ROADMAP

What we cover today

01
Administration
role, servers, duties

02
Processes
ps, top, kill, signals

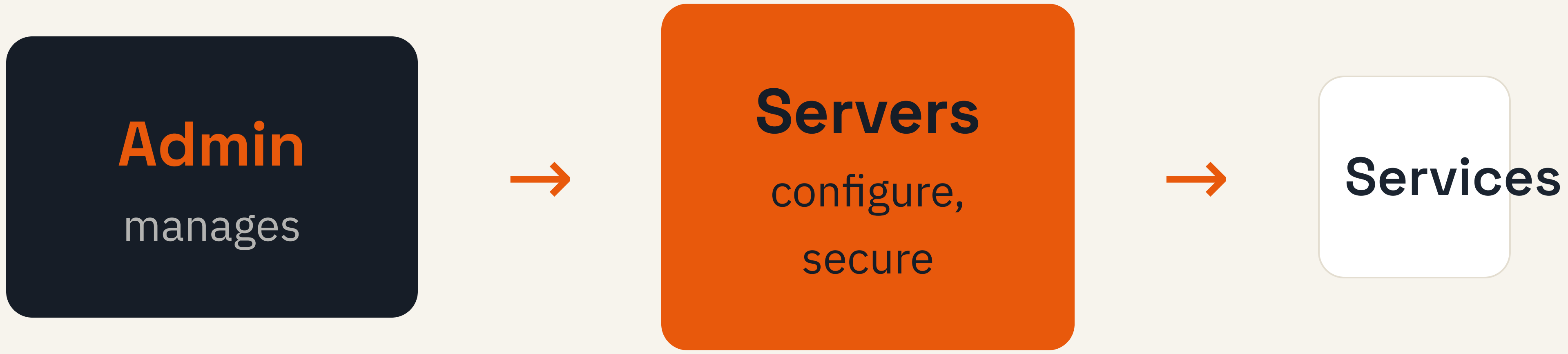
03
Scheduling
cron, crontab, jobs

04
Networking
IP, DNS, ping, ip addr

THE ROLE

What is Linux administration?

Keeping Linux servers healthy, secure, and running the services users depend on.



DAY TO DAY

Responsibilities of an admin

Users

accounts, groups, permissions

Processes

monitor, kill, schedule

Storage

disks, backups, cleanup

Network

IPs, DNS, connectivity

Security

updates, firewall, access

Services

web, database, uptime

PROGRAMS IN MOTION

What is a process?

Program
file on disk

→

Process
running instance

→

Execution

BIRTH TO EXIT

Process lifecycle

New → Ready → Running → Waiting → Terminated

Waiting ↺ Ready while the CPU is busy elsewhere

WHO STARTED WHOM

Parent and child processes

Parent (bash)

↳ forks

Child 1

Child 2

Every process has a `PPID` pointing to its parent.

The parent waits, then reaps the child when it exits.

THE PROCESS TREE

Process hierarchy in Linux

WHERE IT RUNS

Foreground vs background

Foreground

Holds the terminal until it finishes.

```
$ ./build.sh
```

Background

Runs while you keep using the shell.

```
$ ./build.sh &
```

LOOKING INSIDE

Viewing processes

ps

snapshot of processes

top

live, refreshing view

htop

interactive, colourful

SNAPSHOT

The ps command

Lists processes running right now with their PID and command.

terminal

```
imran@ubuntu:~$ ps
PID TTY TIME CMD
2417 pts/0 00:00:00 bash
2980 pts/0 00:00:00 ps
```

SEEING EVERYTHING

Common ps options

aux shows every process for every user.

terminal

```
imran@ubuntu:~$ ps aux
USER PID %CPU %MEM COMMAND
root 1 0.0 0.4 systemd
www 1421 2.1 1.8 nginx
imran 2417 0.0 0.3 bash
```

LIVE VIEW

The top command

A real-time dashboard that refreshes every few seconds.

```
top

top - 14:22:01 up 12 days, load 0.42
Tasks: 142 total, 1 running
%Cpu(s): 7.3 us, 2.1 sy, 90.6 id
PID USER %CPU %MEM COMMAND
1421 www 2.1 1.8 nginx
3902 imran 1.4 4.2 node
2417 imran 0.0 0.3 bash
```

READING TOP

Understanding CPU usage

us

user programs

sy

kernel / system

id

idle, free capacity

High us + low id = the CPU is the bottleneck

RAM AT A GLANCE

Understanding memory usage

```
imran@ubuntu:~$ free -h
total used free
Mem: 7.6G 3.1G 4.5G
Swap: 2.0G 0.0G 2.0G
```

used — held by processes

free — available now

swap — disk used as overflow

FINDING THE CULPRIT

Spotting heavy processes

Sort `ps` by CPU to find what is hogging the server.

terminal

```
imran@ubuntu:~$ ps aux --sort=-%cpu | head
USER PID %CPU %MEM COMMAND
imran 3902 98.7 4.2 node
www 1421 2.1 1.8 nginx
```

STOPPING WORK

The kill command

Sends a signal to a process by its PID asking it to stop.

terminal

```
imran@ubuntu:~$ kill 3902
[asks PID 3902 to terminate]
imran@ubuntu:~$ kill -SIGTERM 3902
[same polite request]
```

MESSAGES TO PROCESSES

The signal concept

kill sends signal → **Kernel** delivers it → **Process**

SIGTERM asks nicely · SIGKILL forces · SIGHUP reloads

THE LAST RESORT

kill -9

Sends SIGKILL. The process cannot ignore it and stops at once.

Use only when a process is frozen — it gets no chance to clean up.

terminal

```
imran@ubuntu:~$ kill -9 3902
[3902 killed immediately]
imran@ubuntu:~$ ps -p 3902
no such process
```

YOUR TURN

Process management exercise

1

List all processes with `ps aux`.

2

Find the top CPU user in `top`.

3

Stop it with `kill`, then verify.

WORK WHILE YOU SLEEP

Why task scheduling matters

Clock
time arrives

→

Scheduled task
triggers

→

Execution

THE SCHEDULER

Introduction to cron

A background service that runs commands automatically at set times.

`crond`

the daemon that checks the clock

`crontab`

your list of timed jobs

HOW CRON FITS TOGETHER

Crontab architecture

User → Crontab → Cron daemon → Script

Edit jobs with `crontab -e`, list them with `crontab -l`

FIVE FIELDS

Crontab syntax

*	*	*	*	*
Minute	**Hour**	**Day**	**Month**	**Weekday**
0–59	0–23	1–31	1–12	0–6

```
30 2 * * * /home/imran/backup.sh
```

READING SCHEDULES

Cron scheduling examples

`0 * * * *`	every hour, on the hour
`30 2 * * *`	every day at 02:30
`0 9 * * 1`	every Monday at 09:00
`*/15 * * * *`	every 15 minutes

ON REAL SERVERS

Real world cron use cases

Backups

copy data every night

Cleanup

delete old logs and temp files

Reports

email stats each morning

Health checks

ping services on a timer

EXAMPLE ONE

Daily backup

Archive a folder to a backup directory every night at 02:00.

crontab -e

```
# minute hour day month weekday
0 2 * * * tar -czf \
  /backup/site.tgz /var/www
```

EXAMPLE TWO

Automated cleanup

Delete log files older than seven days, every Sunday at midnight.

crontab -e

```
# every Sunday at 00:00
0 0 * * 0 find /var/log \
  -name "*.log" -mtime +7 -delete
```

MACHINES TALKING

Introduction to networking

Device
your machine

→

Router
directs traffic

→

Internet

THE CLOUD IS A NETWORK

Why networking matters in cloud

Access

reach servers by IP and DNS

Connect

apps talk to databases and APIs

Debug

find why a service is unreachable

ADDRESSES ON A NETWORK

What is an IP address?

A unique number that identifies a device so data knows where to go.

IPv4

`192.168.1.10`

four numbers, 0–255

IPv6

`2001:db8::1`

much larger space

TWO KINDS OF ADDRESS

Public vs private IP

Public

Reachable from the internet.

`203.0.113.5`

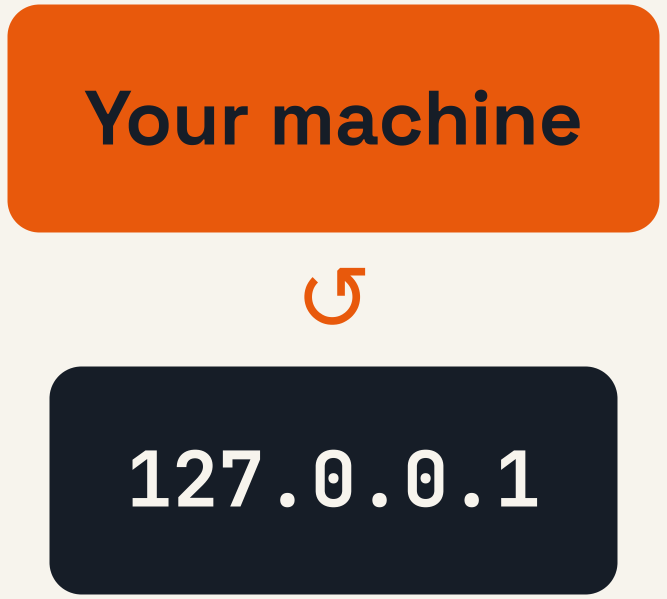
Private

Used inside a local network only.

`192.168.1.10`

TALKING TO YOURSELF

localhost and 127.0.0.1



request loops back, never leaves

`localhost` always points to the machine you are on.

Great for testing a web app before it goes live.

WHO AM I?

The hostname command

Shows the name of the machine you are working on.

bash

```
imran@ubuntu:~$ hostname
ubuntu-server
imran@ubuntu:~$ hostname -I
192.168.1.10
```

IS IT REACHABLE?

The ping command

Sends small packets to test if a host responds, and how fast.

```
bash

imran@ubuntu:~$ ping google.com
64 bytes from 142.250.x.x: time=12.4 ms
64 bytes from 142.250.x.x: time=11.8 ms
3 packets transmitted, 3 received, 0% loss
```

THE INTERNET PHONEBOOK

DNS basics

Name
google.com

→

DNS
looks up the number

→

142.250.x.x

END TO END

What happens when you visit a site

1. You type `google.com`
2. DNS turns the name into an IP address
3. Your request travels to that server
4. The server sends the page back to you

INSPECT INTERFACES

The ip addr command

Lists your network interfaces and the IP addresses assigned to them.

bash

```
imran@ubuntu:~$ ip addr
2: eth0: <BROADCAST,UP>
  inet 192.168.1.10/24
  link/ether 08:00:27:ab:cd:ef
```

A SIMPLE CHECKLIST

Troubleshooting connectivity

`ping 127.0.0.1`

Is my own stack alive?

`ping gateway`

Can I reach the router?

`ping 8.8.8.8`

Is the internet reachable?

`ping google.com`

Is DNS working too?

TRY IT YOURSELF

Networking exercise

1

Find your machine name and IP.

`hostname -I`

2

Ping a public website four times.

`ping -c 4 google.com`

3

List your network interfaces.

`ip addr`

SECTION RECAP

Administration summary

The role

admins keep systems running, secure and backed up

Automation

cron runs jobs on a schedule, no human needed

Mindset

prevent problems before they happen

SECTION RECAP

Process management summary

`ps`	snapshot of running processes
`top`	live view of CPU and memory
`kill`	send a signal to stop a process
`cron`	run jobs automatically on time

SECTION RECAP

Networking summary

`IP address`	unique number for each device
`DNS`	turns names into IP addresses
`ping`	test if a host responds
`ip / hostname`	inspect your own machine

THE BIG PICTURE

Revision mind map

Linux Server Skills

↓

Administration

role · cron · automation

Processes

ps · top · kill · signals

Networking

IP · DNS · ping · ip addr

CHECK YOURSELF

Quiz recap

Q1

Which command shows a live view of CPU usage?

Q2

What does kill -9 do to a process?

Q3

What does DNS translate, and into what?

Q4

Write a cron line for daily 2:30 AM.